

An Analysis of Lightweight Models for Document Image Machine Translation

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Abstract. Document Image Machine Translation (DIMIT) translates text in document images into a target language while preserving reading order. This task is challenging due to complex layouts, multi-column structures, tables, and OCR noise. While large multi-modal models achieve strong results, their high computational cost limits practical deployment, especially in resource-constrained settings. We therefore explore whether lightweight alternatives can offer competitive performance more efficiently. We present a comparative study of lightweight modular pipelines and multi-modal approaches (one- and two-stage) for English-to-Chinese DIMIT. Specifically, we evaluate: (1) lightweight pipelines combining compact document text extraction or reading-order generation models (TrOCR, Mistral OCR, and fine-tuned LayoutLMv3+T5-Small) with MarianMT; (2) large-model pipelines using few-shot prompting; and (3) multi-modal systems, including direct end-to-end generation (Pixtral) and DeepSeek-OCR-2-based variants, to examine trade-offs between simplicity, controllability, and performance. Experiments on the ICDAR 2025 DIMIT benchmark show that fine-tuned lightweight pipelines achieve strong translation quality with far fewer parameters: LM3-T5-V2 $\rightarrow$ MarianMT reaches 0.3627 BLEU, outperforming Gemma-3 $\rightarrow$ MarianMT (0.3225). Pixtral achieves 0.2928 BLEU in the direct setting, remaining below the best modular pipeline, while DeepSeek-OCR-2 reaches 0.0275 BLEU directly but improves to 0.3023 when paired with MarianMT. Overall, our results highlight practical trade-offs between accuracy and computational cost, providing guidance for deploying DIMIT systems under different resource constraints.

Keywords: Document Image Translation · Lightweight Models · Reading Order Recovery · Multi-modal Learning · Neural Machine Translation

1 Introduction

The large and continually increasing volume of digitized documents, such as scanned forms, academic papers, web pages, and historical archives, has created an urgent need for automated systems capable of understanding and translating document images across languages. Document Image Machine Translation (DIMIT) addresses this challenge by combining optical character recognition (OCR), layout understanding, and neural machine translation (NMT) into a unified pipeline that converts source-language document images into target-language text while preserving semantic coherence and reading order.

Unlike standard machine translation, DIMIT faces unique challenges stemming from the visual nature of its input. Document images exhibit complex layouts including multi-column text, tables, figures, headers, footers, and sidebars. Standard OCR systems typically extract text in scan order (left-to-right, top-to-bottom within detected regions), which often differs substantially from the logical reading order intended by document authors. NMT systems are typically pre-trained on well-ordered text, which implicitly constrains the form of the input they can handle reliably. As a result, upstream OCR errors, particularly reading order mistakes, may not only propagate but also be amplified during translation, leading to incoherent outputs [21].

Recent advances in large multi-modal models (LMMs) have demonstrated impressive capabilities in document understanding. Models such as GPT-4V [11,10], Gemini [3], Pixtral [2], and DeepSeek-OCR-2 [17] can process document images directly and generate coherent text outputs. However, these models typically contain tens to hundreds of billions of parameters, making direct on-device or self-hosted deployment expensive or impractical in resource-constrained settings. While cloud-based services can still provide relatively fast inference, they introduce recurring operational costs, dependence on network connectivity, and latency variability, and may be unsuitable for privacy-sensitive or regulated document-processing scenarios. These considerations motivate a fundamental research question: *can lightweight models achieve competitive DIMIT performance at a fraction of the computational and operational cost?*

In this work, we provide a systematic study of *lightweight* approaches for English-to-Chinese DIMIT under realistic resource constraints. Our contributions are threefold. We present:

1. a unified experimental comparison of three paradigms on the ICDAR 2025 DIMIT benchmark: *lightweight modular* reading-order $\rightarrow$ translation pipelines, *large-model* few-shot reading-order text generation pipelines, and *multi-modal* direct and two-stage systems;
2. a strong compact baseline based on LayoutLMv3+T5-Small for reading-order recovery and an analysis of the effect of extended training on both reordering and downstream translation;
3. a multi-modal evaluation that separates OCR/reading-order quality from translation quality via direct vs. hybrid settings, helping attribute failure modes and deployment trade-offs.

2 Related Work

2.1 Document Image Understanding

Document image understanding involves information extraction from visually rich documents, including core sub-tasks such as OCR and layout analysis. LayoutLM [19] pioneered the integration of text, layout, and image information by extending BERT with 2D positional embeddings derived from OCR bounding boxes, demonstrating that spatial information significantly improves document understanding tasks. LayoutLMv2 [18] advanced this paradigm by adding visual features through a multi-modal Transformer architecture, while LayoutLMv3 [5] unified text and image masking during pre-training, achieving strong performance on document understanding benchmarks including form understanding, receipt recognition, and document classification.

Recovering correct reading order from OCR output remains challenging for complex document layouts. Rule-based approaches sort text boxes by geometric coordinates fail on multi-column layouts where this heuristic produces interleaved text from adjacent columns. Learning-based methods train models to predict reading sequences through pairwise classification [16] or sequence generation. LayoutReader [16] formulates reading order detection as a sequence-to-sequence task, using layout-aware representations to generate correctly ordered token sequences. Recent work also explores prompting-based approaches for reading-order recovery, leveraging strong language understanding to infer logical order from contextual cues [1,7].

2.2 Document Image Machine Translation

DIMIT extends document understanding to cross-lingual settings, requiring both layout comprehension and translation capability. Traditional DIMIT systems cascade separate OCR, reading-order recovery, and translation modules, where each component can be optimized independently, but errors propagate through the pipeline. The ICDAR 2025 DIMIT competition [20] provides standardized benchmarks for evaluating such systems on English-to-Chinese translation.

LayoutDIT [21] proposed an end-to-end architecture combining LayoutLMv3 encoding with a multi-step conductive decoder for direct image-to-translation generation, eliminating explicit reading-order recovery stages. Liang *et al.* [8] introduced dynamic multi-pretrained model assembly to weight different expert modules based on document characteristics, improving translation quality on heterogeneous document collections.

Recent LMMs demonstrate strong document understanding capabilities in few-shot or zero-shot settings. Pixtral [2], a multi-modal model released by Mistral AI, processes document images alongside text prompts, generating coherent text outputs without task-specific fine-tuning. DeepSeek-OCR-2 [17] is an OCR-oriented multi-modal model designed for document parsing and text extraction, and is relevant to DIMIT as a strong single-model OCR/reading-order component. Gemma-3 [4] is an open, instruction-tuned multi-modal model family

that supports joint image-text input and demonstrates strong zero-shot and few-shot performance on document understanding tasks.

2.3 Efficient NLP Models

Raffel *et al.* introduced T5 [13], a unified text-to-text framework with models ranging from T5-Small (60M parameters) to T5-11B, where smaller variants provide efficient alternatives for resource-constrained deployment. MarianMT [15] provides translation models for hundreds of language pairs trained on OPUS parallel corpora, with the English-Chinese model containing approximately 78M parameters. TrOCR [6] combines vision and language Transformers for end-to-end text recognition. In this work, we use the TrOCR-Base variant (333M parameters) as a lightweight baseline and adapt it for reading-order recovery through fine-tuning on reading-order supervision.

3 Methodology

We investigate DIMT through three complementary approaches: lightweight modular pipelines combining reading-order handling and translation stages, large-model pipelines using few-shot prompting, and multi-modal systems (direct end-to-end and two-pass prompting variants). Figure 1 illustrates these architectures.

3.1 Task Definition

Given a document image I containing source-language text (English), the DIMT task requires producing target-language text (Chinese) T that accurately translates the document content while preserving logical reading order. The task decomposes into three stages: text extraction where OCR extracts words $W = \{w_1, w_2, \dots, w_n\}$ with bounding boxes $B = \{b_1, b_2, \dots, b_n\}$ from image I ; reading order recovery where extracted words are reordered into the correct reading sequence $W' = \{w_{\pi(1)}, w_{\pi(2)}, \dots, w_{\pi(n)}\}$ with π being the permutation mapping scan order to reading order; and translation where reordered English text W' is translated into Chinese text T . The modular pipeline approach explicitly separates these stages, while multi-modal approaches either perform these stages implicitly in a single pass (direct end-to-end) or decompose them into prompted passes within the same model. In this paper, we distinguish between *OCR text extraction*, *reading-order recovery*, and *translation*. The ICDAR DIMT dataset provides OCR words and bounding boxes, but in our evaluated systems only the LayoutLMv3+T5 pipeline explicitly uses these word-level OCR tokens and bounding boxes as model inputs for layout-aware reading-order recovery. In contrast, TrOCR, Mistral OCR, Gemma-3, Pixtral, and DeepSeek-OCR-2 primarily operate on the full document image and generate text directly (with reading order handled implicitly or through prompting/rule-based processing), after which translation is performed either directly by the same model or by MarianMT in hybrid pipelines.

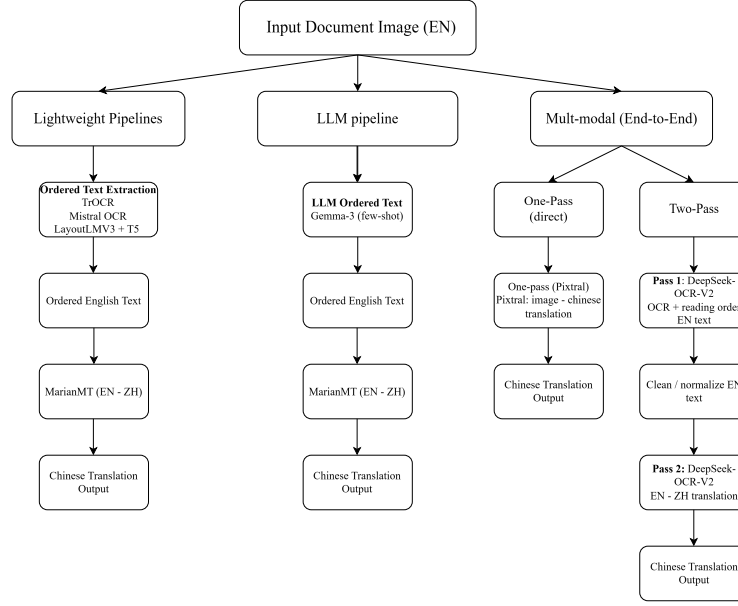


Fig. 1. Overview of evaluated DIMIT approaches: (a) Lightweight modular pipeline with separate reading-order and translation stages; (b) Large-model pipeline with few-shot reading-order; (c) Multi-modal translation approach (direct end-to-end shown; a two-pass DeepSeek-OCR-2 variant is described in the text).

3.2 Lightweight Modular Pipelines

Our primary focus is on lightweight modular pipelines that decompose DIMIT into reordering and translation stages, each handled by compact models optimized for their specific tasks.

Stage 1: Reading Order Recovery. We evaluate four reading-order approaches spanning different model scales and training paradigms.

TrOCR Baseline. We use the standard TrOCR-Base model [6] as an adapted baseline for page-level reading-order text generation. Rather than applying a separate upstream layout/line-detection module or a specialized multi-line/page variant, we fine-tune TrOCR directly on full document images and train it to autoregressively generate the complete page text in reading order. TrOCR combines a Vision Transformer (ViT) encoder with a text Transformer decoder in a VisionEncoderDecoderModel architecture. Originally designed for end-to-end text recognition on images, TrOCR uses a ViT-based encoder initialized from pre-trained vision Transformer weights (e.g., DeiT or BEiT, depending on the TrOCR variant) and a Transformer decoder initialized from pre-trained RoBERTa weights for text generation. The TrOCR-Base variant used here contains 333 M

parameters. While TrOCR was originally pre-trained for text recognition on cropped text-line/word images, we apply it here at page level.

For DIMIT, we adapt TrOCR as a page-level sequence generation baseline that directly generates the document text in correct reading order from the input image. The model learns to produce text in reading order from the raw page image alone, i.e., by relying on spatial and visual patterns implicitly encoded by the TrOCR visual encoder, without explicit layout annotations or a separate layout-analysis model. We fine-tune TrOCR on the ICDAR DIMIT training set.

Training proceeds for 20 epochs with 6,000 sampled images per epoch to ensure layout diversity while maintaining moderate training times. For training, we use the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 4 and FP16 mixed precision. During inference, we generate page-level text outputs on the validation and test sets with batch size 4, and evaluate them against the ground-truth reading-order text using BLEU with uniform 4-gram weights. While TrOCR provides a functional baseline, it relies solely on image features without explicit layout information which limits performance on documents with subtle layout distinctions.

Mistral OCR. Mistral OCR v25.03 [9] provides a lightweight inference-only baseline requiring no fine-tuning or few-shot prompting. Released by Mistral AI in early 2025, this OCR system offers implicit layout understanding through its document processing capabilities.

The OCR output returns structured page objects containing markdown-formatted text with positional information. We apply rule-based reflow using top-to-bottom, left-to-right sorting based on word bounding box coordinates to establish reading order. This geometric sorting approach serves as a reference point for minimal-effort DIMIT: given access to a capable modern OCR system, how well does simple coordinate-based ordering perform? This inference-only baseline serves as a simple lower-bound reference, showing that modern OCR outputs can be reasonably ordered using geometric heuristics. The inference-only nature makes this approach particularly attractive for rapid deployment scenarios where training resources are unavailable.

LayoutLMv3 + T5-Small (LM3-T5). Our primary lightweight pipeline combines LayoutLMv3-Base [5] for multi-modal encoding with T5-Small [13] for sequence generation, creating a powerful yet compact architecture for reading order recovery.

LayoutLMv3 represents the third iteration of Microsoft’s layout-aware document understanding models. Unlike its predecessors, LayoutLMv3 unifies text and image masking during pre-training, learning joint representations of text tokens, layout positions, and document images. The Base variant contains 125.3M parameters and processes three input modalities: document images via a linear patch embedding layer, text tokens via word embeddings, and spatial layout via 2D positional embeddings derived from OCR bounding boxes. The model produces 768-dimensional contextualized representations that encode both semantic and spatial relationships.

T5-Small, part of Google’s Text-to-Text Transfer Transformer family, provides an efficient encoder-decoder architecture with 60.5 M parameters. Pre-trained on the Colossal Clean Crawled Corpus (C4) [13] using a denoising objective, T5 frames all NLP tasks as text-to-text problems, making it naturally suited for generating reordered text sequences. The model expects 512-dimensional input embeddings.

To bridge LayoutLMv3’s output dimension (768) with T5-Small’s input dimension (512), we introduce a learned projection block containing a linear layer, LayerNorm, and GELU activation, totaling 0.4 M parameters. The complete LM3-T5 cascade comprises 186.2 M parameters, positioning it in a substantially lower parameter regime than billion-parameter alternatives.

Model optimization is performed in an end-to-end manner on the reading-order recovery task. For each training sample, LayoutLMv3 processes the document image together with externally provided OCR-extracted words and their bounding boxes (i.e., OCR is not run inside the LayoutLMv3 preprocessing step). The final hidden states of the encoder output are propagated through the projection block, producing embeddings that serve as input for T5-Small’s encoder. The T5 decoder then generates the correctly ordered text sequence autoregressively, supervised by cross-entropy loss:

$$\mathcal{L} = - \sum_{t=1}^{|W'|} \log P(w'_t | w'_{<t}, \text{LM3}(I, W, B)) \quad (1)$$

where LM3 denotes the LayoutLMv3 encoder processing image I , words W , and bounding boxes B .

We train LM3-T5 under two regimens to analyze the impact of training duration. Both configurations use AdamW optimizer with a learning rate of 3×10^{-5} , weight decay 0.01, 10 % linear warmup, gradient clipping at max norm 1.0, batch size 8, and FP16 mixed precision on 4×A100 GPUs. During each epoch we sample 6,000 images from the 300 K training pool to ensure layout diversity. The 30-epoch configuration serves as a shorter training baseline, while the 100-epoch configuration is used as a controlled extended-training setting to quantify the effect of training duration on reordering and downstream translation quality, rather than to claim full convergence.

Stage 2: Translation. All correctly ordered outputs of stage 1 are fed into a shared MarianMT model [15] for English-to-Chinese translation.

MarianMT. MarianMT represents Helsinki-NLP’s contribution to the OPUS-MT project, providing neural machine translation models for hundreds of language pairs trained on OPUS parallel corpora. The English–Chinese MarianMT checkpoint (Helsinki-NLP/opus-mt-en-zh)¹ [15] contains approximately 77.9 M parameters in a standard encoder-decoder Transformer architecture, offering efficient translation suitable for integration into resource-constrained pipelines.

¹ <https://huggingface.co/Helsinki-NLP/opus-mt-en-zh>

We fine-tune MarianMT for 75 epochs on DIMT training pairs, using the ground-truth reordered English document text as source and the corresponding Chinese reference translation as target. Training employs a batch size of 8 with gradient accumulation 2 (effective batch size 16), learning rate 2×10^{-5} , 500 warmup steps, and a maximum source/target sequence length of 512, used as a fixed training/inference cap for efficiency and memory stability. We use FP16 mixed-precision training with dynamic padding.

During inference, we use a batch size of 32 with beam search decoding with 5 beams and early stopping. Generated Chinese text is post-processed with Jieba segmentation [14] before BLEU evaluation, because Chinese text does not contain whitespace-delimited words. We use Jieba in its standard (non-exhaustive) segmentation mode, which produces a single tokenization of the text rather than enumerating multiple possible segmentations.

3.3 Large-Model Pipeline Comparator

To contextualize lightweight pipeline performance, we compare against a large-model reordering approach representing the capability frontier without strict resource constraints.

Gemma-3 Few-Shot. Gemma-3 [4], developed by Google DeepMind, represents the third generation of the Gemma family of lightweight yet capable open models. The 12.2 B parameter variant we employ offers strong multi-modal understanding through image-conditioned text generation, having been trained on diverse web documents, code, and mathematical content. It performs reading order recovery through few-shot prompting, requiring no task-specific fine-tuning.

Our few-shot prompt design includes a system message establishing the OCR assistant role: “You are an OCR assistant. For every provided image, extract the text in correct reading order. Return your answer as a plain text string with no extra commentary.” We then provide 13 image-to-ordered-text examples as in-context demonstrations each showing a document image (encoded as base64 data URI) paired with its ground-truth reading order. These examples are selected to cover diverse layouts including single-column text, multi-column documents, and tables. The final user message presents the target image with the instruction to extract ordered text. Inference uses greedy decoding with a maximum length of 500 tokens.

Reordered English text from Gemma-3 feeds into the same fine-tuned MarianMT model used by lightweight pipelines for consistent translation comparison. This demonstrates the power of large pretrained models to generalize to new tasks with minimal adaptation, though inference requires substantially more computational resources than lightweight alternatives.

3.4 One- and Two-Stage Multi-modal Systems

A key contribution of this work is the investigation of multi-modal approaches that reduce or bypass explicit modular reordering and translation pipelines, using

either a single direct pass (Pixtral) or a controlled two-stage decomposition (DeepSeek-OCR-2). These systems offer potential simplification by eliminating multi-stage pipelines, though they present different trade-offs in controllability and interpretability.

Pixtral End-to-End. Pixtral [2], developed by Mistral AI, represents a natively multi-modal model from the Mixtral family with 12.7B parameters. Unlike models that retrofit vision capabilities onto existing language models, Pixtral was designed and trained from scratch to process interleaved image and text inputs, making it particularly well-suited for document understanding tasks requiring joint visual and textual reasoning.

The architecture processes document images as base64 data URIs alongside text prompts, generating coherent text outputs without requiring separate OCR or layout analysis stages. For DIMT, we investigate whether Pixtral can directly produce Chinese translations from English document images, combining OCR, reading-order handling, and translation into a single inference pass.

In the zero-shot configuration, we provide only the document image and a translation instruction without examples. The system prompt establishes the model’s role: “You are a document translation assistant. Your task is to extract all text from document images and translate it to Chinese while preserving the correct reading order.” The user prompt requests: “Please look at this document image carefully. Extract all the text in the correct reading order and translate it completely to Chinese. Output only the Chinese translation, nothing else.” We use greedy decoding with a maximum of 2048 tokens and a temperature of 0.0 for deterministic outputs.

The few-shot configuration provides 3–5 exemplar document images with their corresponding Chinese translations to improve output consistency. Examples cover diverse layouts (single-column, multi-column, tables) and demonstrate the expected input-output format. Each example shows the document image followed by the complete Chinese translation, establishing the mapping the model should learn to replicate. Pixtral inference uses the pixtral-12b-2409 model.

We additionally fine-tune Pixtral on the ICDAR DIMT training split for image-to-ordered-English text generation. The model is optimized using an autoregressive cross-entropy objective under teacher forcing for 100 epochs, following the same ordered-English supervision used in the lightweight pipelines. During inference, the generated English text is translated using MarianMT, enabling direct comparison with the other hybrid approaches.

DeepSeek-OCR-2 Two-Stage Pipeline. DeepSeek-OCR-2 [17],

developed by DeepSeek AI, is an OCR-oriented multi-modal model that we use in a two-pass prompting pipeline for DIMT. This makes DeepSeek-OCR-2 particularly interesting for document understanding tasks requiring both visual comprehension and text generation.

Initial experiments with direct image-to-Chinese translation revealed inconsistent results, particularly for the English-to-Chinese language pair where the

model’s training distribution may provide uneven coverage. To address this limitation while still leveraging DeepSeek-OCR-2’s document OCR capabilities without external modules, we implement a two-stage approach that decomposes the DIMIT task into sequential sub-problems.

Stage 1: OCR and Reading-Order Text Generation. The document image is input with a carefully designed prompt: “You are a document OCR assistant. Look at this document image and extract all the text in the correct reading order. Consider the document layout carefully—handle multi-column text, tables, and other structures appropriately. Output only the extracted English text in reading order, nothing else.” This pass leverages DeepSeek-OCR-2’s OCR/document parsing capabilities to process complex layouts and produce correctly ordered English text, which is saved to intermediate JSON storage for inspection and subsequent processing.

To improve robustness, we first request plain-text OCR output and use a fallback prompt when the model returns grounded markup or HTML-like table output. We then normalize the response into a plain paragraph by stripping markup and flattening table content before translation.

In addition, we evaluate a hybrid setting where DeepSeek-OCR-2 is used only for OCR/reading-order text generation and the resulting English text is translated using the same fine-tuned MarianMT model as in the modular pipelines.

Stage 2: Translation. The cleaned English text from Stage 1 is translated to Chinese using a second prompted DeepSeek-OCR-2 inference call. In our implementation, this second pass uses the same image-conditioned interface (with a translation-only prompt), while the translated content is conditioned on the cleaned English text produced in Stage 1. The prompt instructs the model: “Translate the following English text to Chinese. Maintain the paragraph structure and produce a natural, fluent Chinese translation. Output only the Chinese translation.” This decomposition simplifies the translation objective by providing a cleaned intermediate English text sequence before the second-pass generation.

This two-stage design offers several advantages despite requiring two inference calls per document. Task decomposition allows the model to focus on one complex task at a time, potentially improving quality on both sub-tasks. Error isolation enables identification of whether failures originate from text extraction or translation, facilitating targeted debugging. Quality control permits inspection of intermediate English output for correctness before translation proceeds.

3.5 Evaluation Metrics

We employ document-level BLEU [12] as our primary evaluation metric, computed separately for reading-order generation and translation tasks. Reading-order BLEU compares ordered English output against ground-truth reading order at the token level using NLTK corpus BLEU with uniform 4-gram weights and smoothing. Translation BLEU compares final Chinese output against reference translations at the character level after Jieba segmentation. We also report parameter counts,

training requirements, and qualitative inference cost assessments. For the principal translation results, we additionally report 95% bootstrap confidence intervals computed over validation examples to quantify uncertainty in the reported BLEU scores.

4 Experiments

4.1 Dataset

We conduct experiments on the ICDAR 2025 DIMT Track 1 benchmark [20], which provides 300,000 document images for training containing web documents and academic articles, each with OCR-extracted words with bounding boxes, ground-truth reading order, and Chinese reference translations. The validation and test sets each contain 1,000 images with complete annotations. The dataset includes diverse document types: web documents such as news articles, blog posts, and product pages; academic articles with two-column layouts, equations, tables, and figures; and varying layout complexity from single-column to multi-column to mixed layouts with embedded elements. Due to computational constraints, we sample 6,000 images per epoch during training, ensuring layout diversity through random selection while maintaining tractable training times. Since official ICDAR DIMT test-set annotations were not publicly released and the Codalab evaluation server was unavailable, all reported results are computed on the validation set.

4.2 Implementation Details

LM3-T5 and TrOCR use $4\times$ NVIDIA A100 40GB GPUs with mixed-precision FP16 for training. MarianMT fine-tuning runs on a single A100 GPU. Large model inference for Gemma-3 and Pixtral uses cloud API endpoints, while DeepSeek-OCR-2 is invoked through Hugging Face. DeepSeek-OCR-2 uses a 3B-parameter MoE decoder with about 500M active parameters [17] and is run in BF16 following the official inference setup². Implementation uses PyTorch 2.0+, HuggingFace Transformers, and standard NLP libraries including NLTK and Jieba. All random seeds are fixed for reproducibility. Table 1 summarizes the key hyperparameters for each model.

4.3 Results: Modular Pipelines

Table 2 summarizes the performance of modular pipelines on the validation set. We report both the English reading-order BLEU and the final Chinese translation BLEU after MarianMT. The TrOCR baseline establishes the performance floor for fine-tuning approaches, while Mistral OCR with rule-based reflow provides a strong inference-only baseline. Notably, the downstream translation BLEU closely aligns with the reading-order BLEU. This correlation confirms that recovering the correct text reading order is the primary bottleneck for DIMT.

² See <https://huggingface.co/deepseek-ai/DeepSeek-OCR-2>.

Table 1. Training and inference hyperparameters for evaluated models. For MarianMT, we use batch size 8 with gradient accumulation 2 (effective 16).

Model	Params	Epochs	LR	Batch size	Precision
TrOCR	333M	20	1×10^{-4}	4	FP16
LM3-T5-V1	186M	30	3×10^{-5}	8	FP16
LM3-T5-V2	186M	100	3×10^{-5}	8	FP16
MarianMT	78M	75	2×10^{-5}	8	FP16
Gemma-3	12.2B	–	–	–	BF16
Pixtral	12.7B	–	–	–	BF16
DeepSeek-OCR-2	3B	–	–	–	BF16

Table 2. Modular pipeline results on the DIMT validation set. Translation BLEU is reported with 95% bootstrap confidence intervals for key systems.

Reordering Model	Params	Setup	Reorder BLEU	Trans. BLEU
TrOCR (baseline)	333M	Fine-tuned (20 ep)	0.3054	0.1946
Mistral OCR + rule reflow	<1B	Inference-only	0.4832	0.2393
LM3-T5-V1	186M	Fine-tuned (30 ep)	0.6060	0.2766
LM3-T5-V2	186M	Fine-tuned (100 ep)	0.8092	0.3627 [0.3519, 0.3731]
Gemma-3 (few-shot)	12.2B	13 examples	0.7178	0.3225 [0.3113, 0.3334]

LM3-T5 shows substantial gains from extended training. Increasing the training from 30 to 100 epochs improves the reordering BLEU by 33.5% relative (from 0.6060 to 0.8092), which leads to a 31.1% relative increase in the translation BLEU (from 0.2766 to 0.3627). Furthermore, the fully trained LM3-T5 outperforms the 12.2B-parameter Gemma-3 few-shot baseline. Despite using only 1.52% of the parameters of Gemma-3, our specialized pipeline achieves a higher translation BLEU (0.3627 vs. 0.3225). This demonstrates that task-specific fine-tuning of small models is highly efficient and effective for layout-sensitive tasks.

4.4 Results: Multi-modal Systems

We evaluate the multi-modal systems across three configurations: (i) *direct* multi-modal translation (image $\rightarrow$ Chinese), (ii) *hybrid* pipelines using a multi-modal model for English OCR and reading-order recovery, followed by MarianMT for translation, and (iii) a *diagnostic* analysis of DeepSeek-OCR-2 to isolate reading-order quality (English BLEU) from translation quality Trans. BLEU).

Practical note on Pixtral few-shot direct evaluation. Due to API rate limits and resource constraints during extensive testing, we could not consistently evaluate Pixtral in the few-shot direct translation setting across the full validation set. To still assess its utility for DIMT, we evaluate Pixtral within the hybrid setting, using it solely to generate the ordered English text for the MarianMT translator.

The hybrid and diagnostic results in Table 4 help isolate the specific failure modes of unified models. DeepSeek-OCR-2 achieves a strong English reading-order

Table 3. Direct multi-modal translation BLEU on the DIMT validation set (image $\rightarrow$ Chinese).

System	Params	Setting	Stages	Trans. BLEU
Pixtral	12.7B	Zero-shot (direct)	1	0.2928
DeepSeek-OCR-2	3B	Two-stage prompted (direct multi-modal)	2	0.0275
LM3-T5-V2 (100 ep) $\rightarrow$ MarianMT	264M	Fine-tuned modular baseline	2	0.3627

Table 4. Hybrid and diagnostic multi-modal results on the DIMT validation set. Translation BLEU is reported with 95% bootstrap confidence intervals for key systems.

System / Pipeline	Passes	English BLEU	Trans. BLEU	Type	Note
Pixtral (few-shot) $\rightarrow$ MarianMT	2	0.5536	0.2894	Hybrid	Pixtral used for OCR/ordered English text
Pixtral (fine-tuned, 100ep) $\rightarrow$ MarianMT	2	0.6336	0.3037 [0.2903, 0.3173]	Hybrid	Pixtral used for OCR/ordered English text
DeepSeek-OCR-2 (two-pass prompted)	2	0.6429	0.0275	Direct	Same model used for OCR + translation
DeepSeek-OCR-2 (prompted) $\rightarrow$ MarianMT	2	0.6429	0.3023 [0.2890, 0.3152]	Hybrid	DeepSeek-OCR-2 used for OCR/ordered English text

score (0.6429 BLEU) but fails at direct translation to Chinese (0.0275 BLEU). However, when we use DeepSeek-OCR-2 only for English OCR and offload the translation to MarianMT, the translation score jumps to 0.3023 BLEU. This confirms that the model successfully understands document layout and text but lacks the specific cross-lingual translation capabilities required for this task.

5 Discussion

5.1 Training Dynamics and Deployment Trade-offs

While our results demonstrate the efficacy of lightweight modular pipelines, analyzing their training dynamics reveals important practical trade-offs. As summarized by the reordered/translation BLEU progression from LM3-T5-V1 (30 epochs) to LM3-T5-V2 (100 epochs) and visualized in Figure 2, LM3-T5 benefits strongly from extended fine-tuning. However, longer training is necessary to surpass the large-scale few-shot baseline: LM3-T5-V2 exceeds Gemma-3 $\rightarrow$ MarianMT in translation BLEU despite using far fewer parameters.

The improvement rate decreases after 80 epochs, suggesting that 100 epochs is a practical operating point. While Gemma-3 enables training-free prototyping, fine-tuning LM3-T5 is more suitable for high-volume DIMT because it improves quality while reducing long-term inference cost.

Our direct multi-modal experiments assess whether a unified approach can replace modular pipelines. As shown in Table 3, the fine-tuned modular baseline (LM3-T5-V2 $\rightarrow$ MarianMT) still significantly outperforms the unified models.

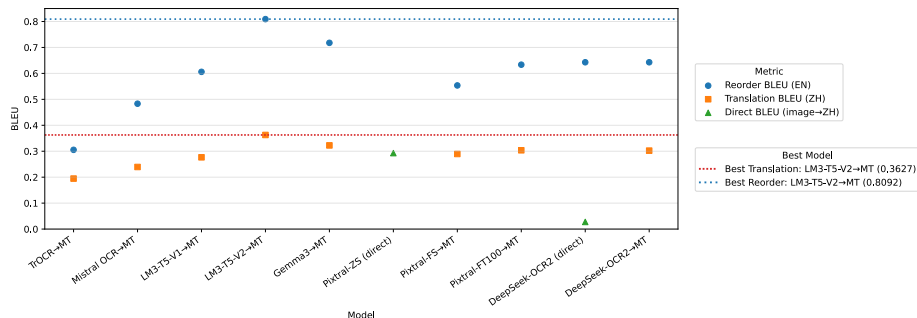


Fig. 2. Scatter plot of BLEU scores across evaluated systems. For each model, we report English reading-order BLEU (reading-order), Chinese translation BLEU (translation), and direct image-to-Chinese BLEU for end-to-end multi-modal runs (direct). Horizontal dotted lines indicate the best observed reading-order and translation scores.

Table 5. Resource-accuracy trade-offs for DIMT approaches.

Approach	Params	Train	Infer.	Trans. BLEU
TrOCR → MarianMT	411M	Low	Low	0.1946
Mistral OCR → MarianMT	<1B	None	Low	0.2393
LM3-T5-V1 → MarianMT	264M	Medium	Low	0.2766
LM3-T5-V2 → MarianMT	264M	High	Low	0.3627
Gemma-3 → MarianMT	12.3B	None	High	0.3225
Pixtral (zero-shot)	12.7B	None	High	0.2928
Pixtral (few-shot) → MarianMT	12.8B	None	High	0.2894
Pixtral (100) → MarianMT	12.8B	High	High	0.3037
DeepSeek-OCR-2 (two-pass)	3B	None	High	0.0275
DeepSeek-OCR-2 (prompted) → MarianMT	3.1B	None	High	0.3023

Figure 2 provides a compact comparison of reordering, translation, and direct multi-modal BLEU across all evaluated systems, highlighting that improved English reading-order recovery generally correlates with higher downstream translation quality.

Table 5 quantifies these resource requirements and performance outcomes across all evaluated approaches. For example, fully training LM3-T5-V2 requires a moderate, one-time investment of approximately three to four days on four A100 GPUs. This upfront hardware cost is quickly offset by the 264M-parameter model’s ability to run low-cost inference indefinitely. This stark contrast with massive models demonstrates that for DIMT, task-specific training and architectural design are significantly more impactful than raw model scale. Furthermore, inference-only baselines remain practically viable; Mistral OCR combined with rule-based reflow achieves a competitive 0.2393 translation BLEU without any training overhead.

5.2 Error Analysis

A qualitative review of the system outputs reveals distinct, recurring failure modes across the different pipelines. Reading-order errors frequently occur when complex multi-column layouts with varying widths cause models to incorrectly interleave text from adjacent columns. Similarly, tables with merged cells often result in scrambled text, and floating elements such as figures or sidebars are frequently misplaced within the reading order. Crucially, these reordering errors propagate heavily into the translation stage. This error cascade often produces Chinese text that is grammatically correct but semantically meaningless.

Translation-specific errors also emerge independently of layout issues. These manifest when models struggle with domain-specific vocabulary or transliterate proper names inconsistently. Additionally, the Jieba segmentation process occasionally introduces boundary errors on compound words.

End-to-end multi-modal systems suffer from their own unique failure modes. These include language mixing, formatting inconsistencies (especially when operating without few-shot examples), and the hallucination of content not present in the source document. Furthermore, context length limitations often lead to truncated outputs for longer documents.

Our multi-modal experiments show that end-to-end performance can be bottlenecked by factors beyond OCR accuracy. Specifically, an OCR-oriented multi-modal model may extract English text and reading order very well, yet fail almost completely at direct English-to-Chinese translation. This underscores the importance of evaluating direct multi-modal performance alongside an intermediate English OCR diagnostic. When necessary, utilizing a hybrid pipeline where the multi-modal model handles OCR and a dedicated model like MarianMT handles translation effectively separates and mitigates these distinct failure modes.

6 Conclusion

In this paper, we addressed Document Image Machine Translation under realistic resource constraints. While recent large multi-modal models offer powerful capabilities, their scale makes them expensive to deploy and unsuitable for privacy-sensitive or offline environments. Through systematic evaluation, we showed that lightweight, modular pipelines can match or exceed the performance of large-scale few-shot models when provided with sufficient task-specific training.

To conclude, we revisit the three study goals from Section 1.

(1) *Paradigm comparison.* Across modular, few-shot, and multi-modal systems, modular reading-order→translation pipelines are the most reliable option and can outperform large few-shot baselines with task-specific training.

(2) *Effect of extended training.* Extended fine-tuning of the compact LM3-T5 baseline improves reading-order recovery and downstream translation quality.

(3) *Direct vs. hybrid multi-modal failure attribution.* Direct vs. hybrid multi-modal evaluation shows that OCR-oriented models can produce strong ordered English but weak direct Chinese translation. Pairing them with MarianMT mitigates this gap and clarifies the bottleneck.

Extending LM3-T5 fine-tuning to 100 epochs yields strong reading-order recovery, enabling the 264M-parameter pipeline to outperform the 12.2B-parameter Gemma-3 in Chinese translation BLEU. The multi-modal analysis further shows that OCR-oriented models can extract text and layout effectively but still struggle with direct cross-lingual translation, supporting hybrid or modular designs.

These trade-offs provide practical deployment guidance: LM3-T5 (100 epochs) with MarianMT offers the best quality-cost balance, Gemma-3 with MarianMT supports rapid prototyping, Mistral OCR with rule-based reflow provides an inference-only option, and multi-modal systems such as Pixtral (direct end-to-end) or DeepSeek-OCR-2 (two-pass) offer simpler but less reliable alternatives.

Future work includes extension of these architectures to additional language pairs, the investigation of knowledge distillation techniques to further compress the models, and a detailed per-layout-type analysis to better identify and prevent specific structural failure modes.

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Code Availability Code for this work is publicly available in the project repository: <https://github.com/Abantika02/ICDAR-2026-An-Analysis-of-Lightweight-Models-for-Document-Image-Machine-Translation>.

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